Agent-based Diamond–Dybvig with cultural heterogeneity

Who runs, and how a run spreads: an agent-based test of the theory

Walter Stover · George Mason University · 26 October 2026

The puzzle: who runs?



CryptoSlate, 12 October 2026, on the Financial Stability Board's note to the G20. Screenshot, cryptoslate.com.

Who runs? Luck of the draw, and location

Variation in the signal

- Goldstein–Pauzner: every depositor gets a noisy draw of the same signal
- The draw breaks common knowledge and selects the equilibrium
- Who runs: whoever drew low
- Everyone decides at once, so there is no first

Variation in location

- Networks and media decide who hears what, from whom, when
- Kelly–Ó Gráda, Iyer–Puri, Cookson et al.
- Who runs first: whoever sits nearest the source

With two signals, a third question appears

Which signal does a depositor act on, and in what proportion?

Latent in the canonical models. Diamond–Dybvig has no signal; Morris–Shin and Goldstein–Pauzner have one. The question cannot arise.

Askable once a public signal is added. Morris–Shin (2002), Hellwig (2002), Angeletos–Werning (2006) put a public signal next to the private one.

The question a panic poses. A depositor in a run holds two things that disagree: what they see and what they are told.

What panics look like

Panics are open disagreements between the public line and private evidence, and depositors split on which to trust.

Celsius, 2022. Official posts insisting the platform was sound, while CEL fell and withdrawals accelerated.

Silicon Valley Bank, 2023. A call asking clients to stay calm, against a Twitter feed saying the opposite.

Northern Rock, 2007. A Bank of England support announcement, against queues outside the branches.

In each case the interesting fact is who moved on which signal.

Cultural economics: a source of stable weights

Culture supplies a weight that differs across depositors, is set before the run, and does not move with the information environment.

Transmitted, not chosen. Cultural values pass largely unchanged from parents to children (Guiso, Sapienza, Zingales; Bazzi et al.). The weight is a disposition, not a calculation.

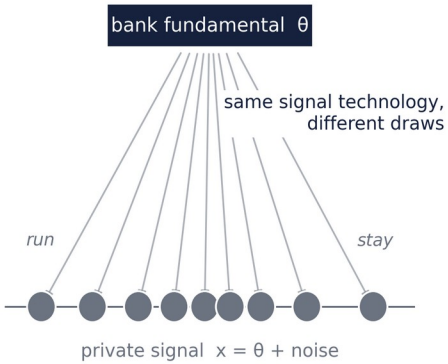
Individualism is the right dimension. Self-reliance against deference to the group (Hofstede) is the private-against-social weighting the model already contains (Gorodnichenko–Roland).

Measurable per depositor. Given names carry it (Twenge et al.), so the weight can be observed, not assumed.

The ladder: signal, location, interpretation

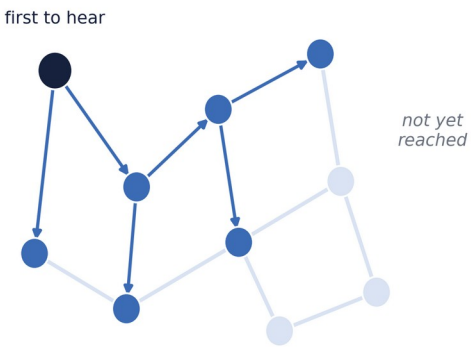
1 Signal

one fundamental, one noisy draw each



2 Location

who hears it, from whom, when

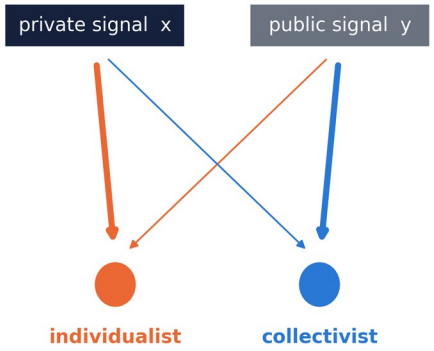


networks and media: Kelly–Ó Gráda, Iyer–Puri, Cookson et al.

3 Interpretation

the same two signals, different weights

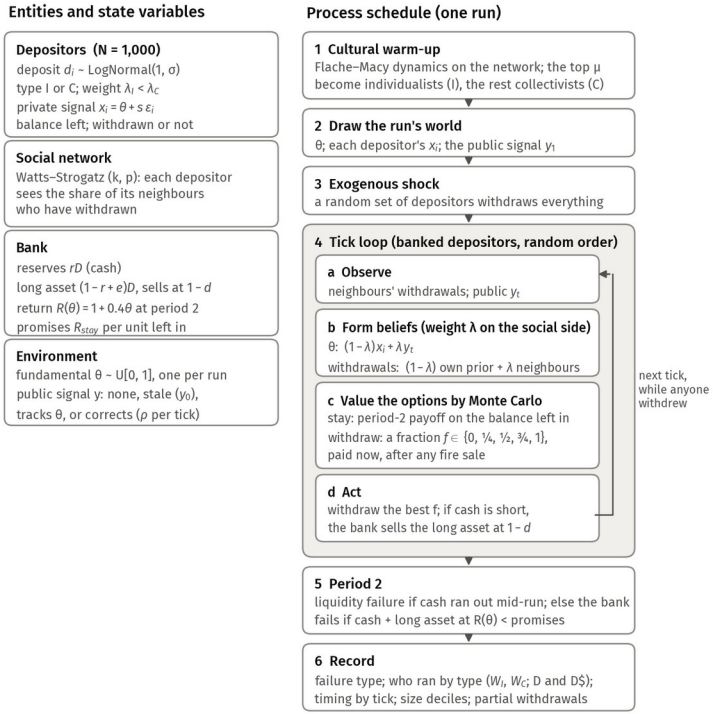
culture sets the weight



leans on what it sees leans on the public line

Rungs 1 and 2 are the literature. **Rung 3 is this paper:** the only one that orders depositors by a trait, and reverses when the public line is too optimistic or too pessimistic.

The model in one picture



1,000
depositors on a small-world network, one bank

1%
cap on any one deposit; sizes log-normal (σ 2.5) below it

CARA 0.1
risk aversion; partial withdrawals allowed

250
runs per cell: 25 initialisations × 10 draws

- One bank that sells its long asset at a discount when cash runs short (a fire-sale bank)
- Each tick a depositor sees neighbours' withdrawals and the public news, blends them with a private signal, and withdraws if that pays
- A public signal can be absent, accurate, or outdated
- Outcomes: D = individualists' minus collectivists' withdrawal share (those not hit by the shock); P(fail) = share of runs that fail

Individualists and collectivists: who listens to whom

Individualists

- Trust their own information about the bank
- Put little weight on what neighbours do or on public news
- Weight on the social side: $\lambda = 0.1$ to 0.3
- Act on what they see themselves

Collectivists

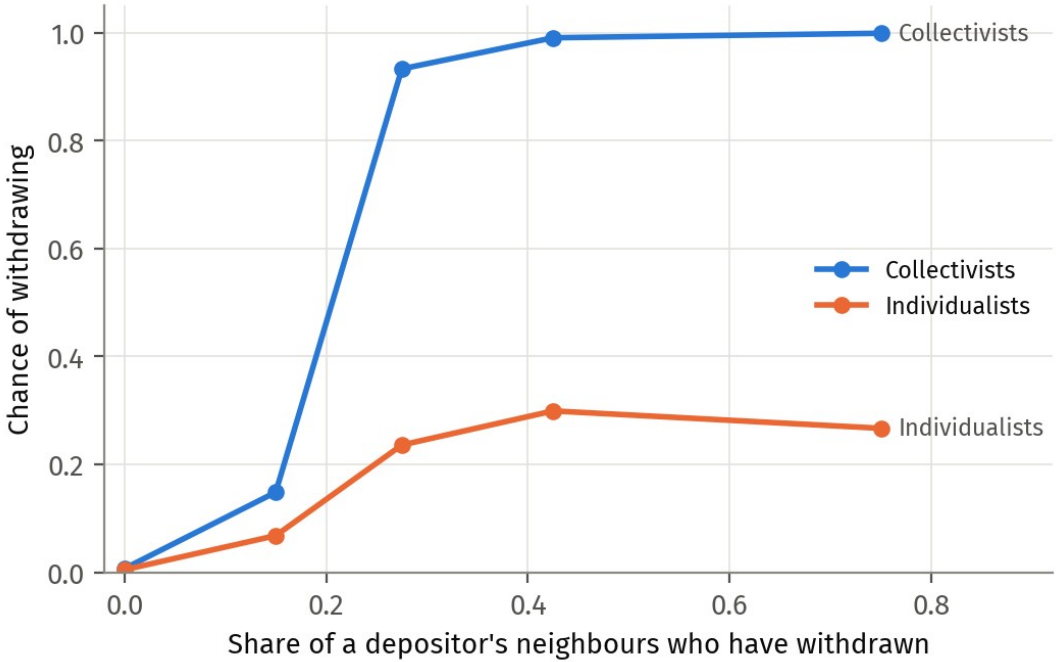
- Watch what neighbours do and what the public news says
- Put little weight on their own information
- Weight on the social side: $\lambda = 0.5$ to 0.9
- Move when others move

Three pairs (λ individualist, λ collectivist) in the figures

(0.1, 0.9) wide gap **(0.2, 0.8)** medium **(0.3, 0.5)** narrow

The share of individualists μ is set by a cultural warm-up on the network. **Orange = individualists**, **blue = collectivists** throughout.

Collectivists herd



Chance of withdrawing, given the share of a depositor's neighbours who have already withdrawn. λ pairs pooled; reference bank.

0.15 to 0.93

collectivists, once 20–35% of their neighbours are out

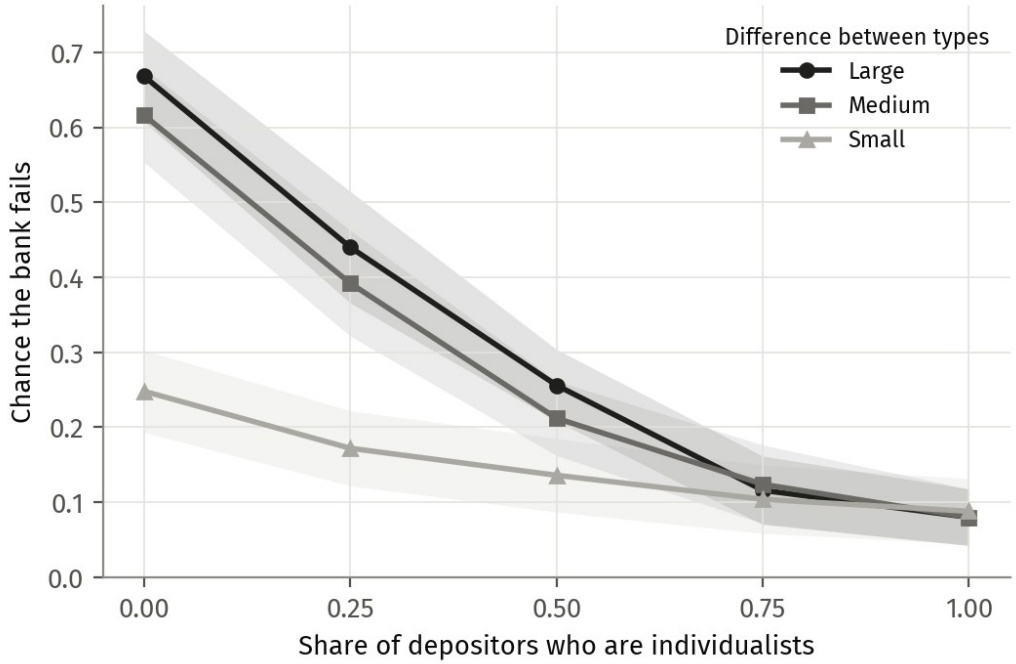
0.25 to 0.30

where individualists level off, whatever their neighbours do

+0.58

difference in slopes (collectivists 0.80, individualists 0.22), t = 16.7

More individualists, fewer failures



Chance the bank fails, by share of individualists μ, one line per λ pair. Includes the pure populations (μ 0 and 1).

0.67 to 0.08

P(fail) from no individualists to all, wide λ gap

0.25 to 0.09

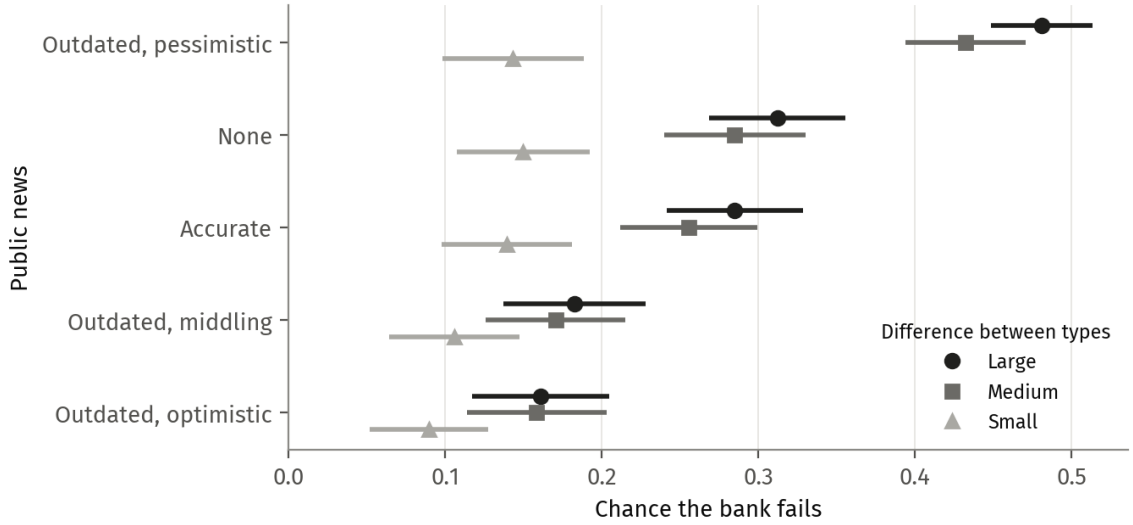
the same, narrow λ gap: it matters most when collectivists dominate

Monotone

No interior peak at any mix. The lines converge once individualists are 75% or more.

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Outdated news cuts both ways



Chance the bank fails, by the kind of public news, dot and CI per λ pair. All mixes μ. Numbers at right: wide λ gap.

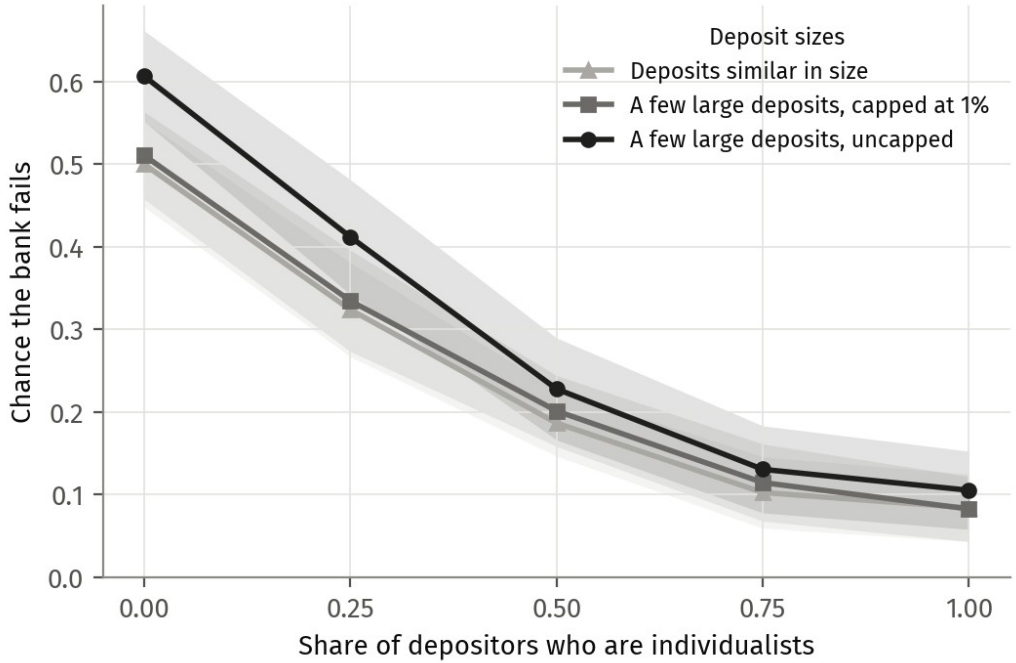
0.48
outdated, pessimistic news

0.31
no public news

0.29
accurate news

0.16
outdated, optimistic news

Capping the largest deposits



Chance the bank fails by share of individualists, for three deposit-size distributions. λ pairs pooled.

0.61 to 0.11

uncapped: a few very large deposits (σ 2.5)

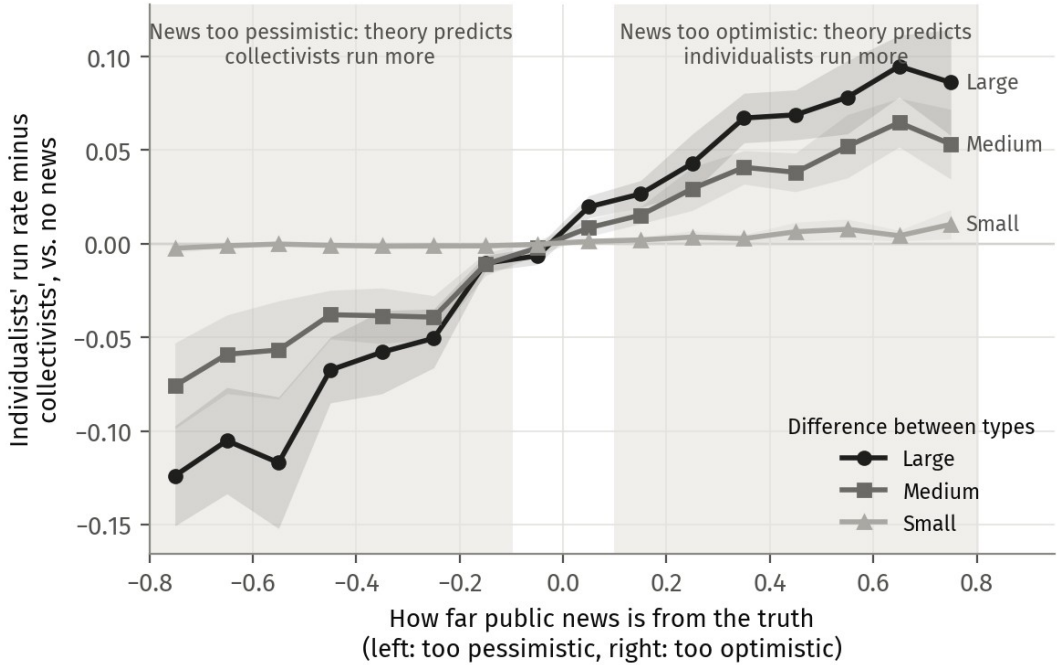
0.51 to 0.08

capped: no deposit above 1% of the total (the reference bank)

0.50 to 0.08

similar sizes (σ 1). The μ gradient is the same in all three.

Does the model hold up?



D relative to no news, against how far the public news sits from the truth. Shaded: where the theory predicts the sign.

Pre-registered tests

G1′ optimistic news: individualists run more +0.033

G4′ pessimistic news: collectivists run more −0.031

G2a′ the effect grows with the λ gap +0.093

G3 sorting with no news: counts against (one pair wrong-signed)

3 of 4

confirmatory tests support the theory, on all three versions of the bank

Takeaways and next steps

1	**Sorting follows the news**	Optimistic public news makes individualists run relatively more; pessimistic news, collectivists. 3 of 4 pre-registered tests pass, in all three bank versions.
2	**Collectivists herd**	Their chance of withdrawing jumps from 0.15 to 0.93 once 20–35% of neighbours are out; individualists level off near 0.3.
3	**More individualists, fewer failures**	P(fail) falls from 0.67 to 0.08 as the individualist share goes from 0 to 1. Monotone; no interior peak.
4	**Stale news cuts both ways**	A stale optimistic signal halves failure, a stale pessimistic one raises it by half. Accurate news barely moves it.
5	**Capping the largest deposits**	A 1% cap brings failure down to the no-concentration bank's level; the culture gradient is unchanged.
6	**Not yet matched**	How Celsius's largest depositors exit: one-shot share 0.35 there, 0.93 or more in the model.

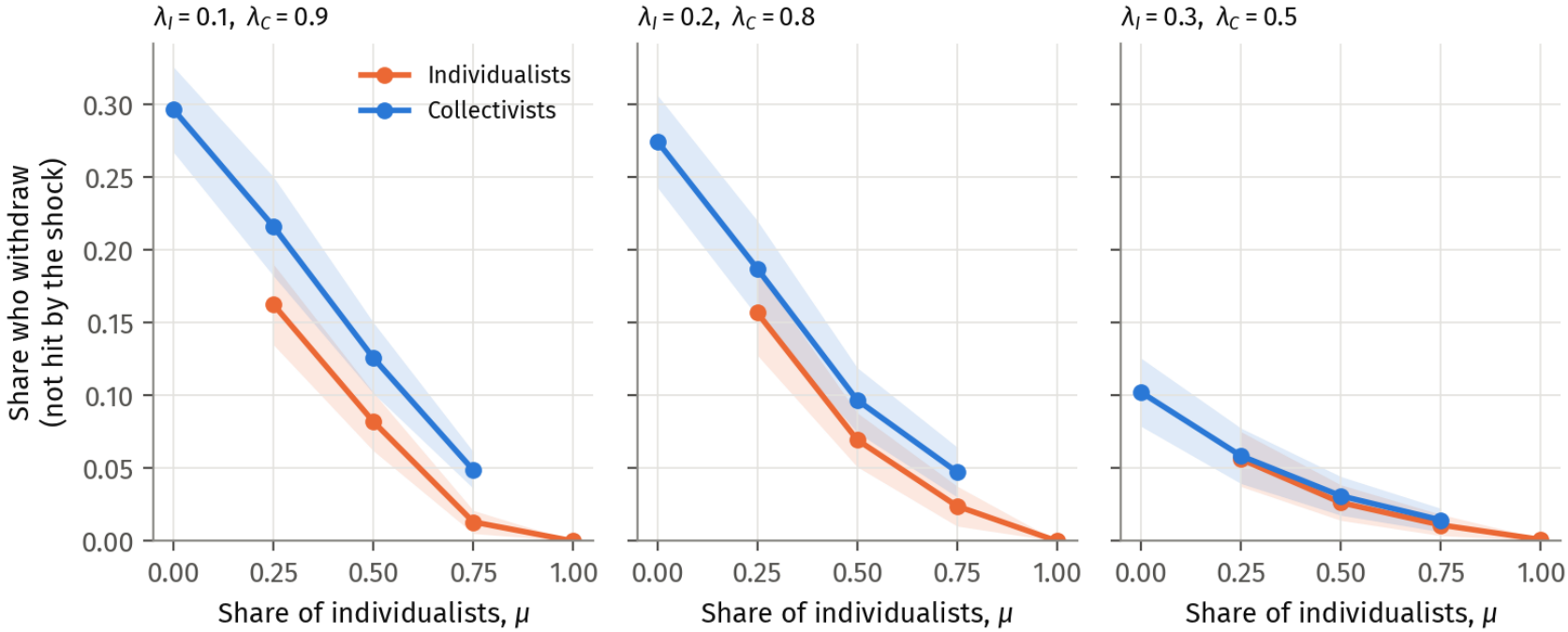
Next steps

- Super-individuals, to match how the largest depositors exited Celsius
- Deposit insurance and the reserve ratio on the fire-sale bank
- Network topology and learning
- Full sweep: stale public signal plus no-signal baseline, CARA 0.1, σ reported

Backup

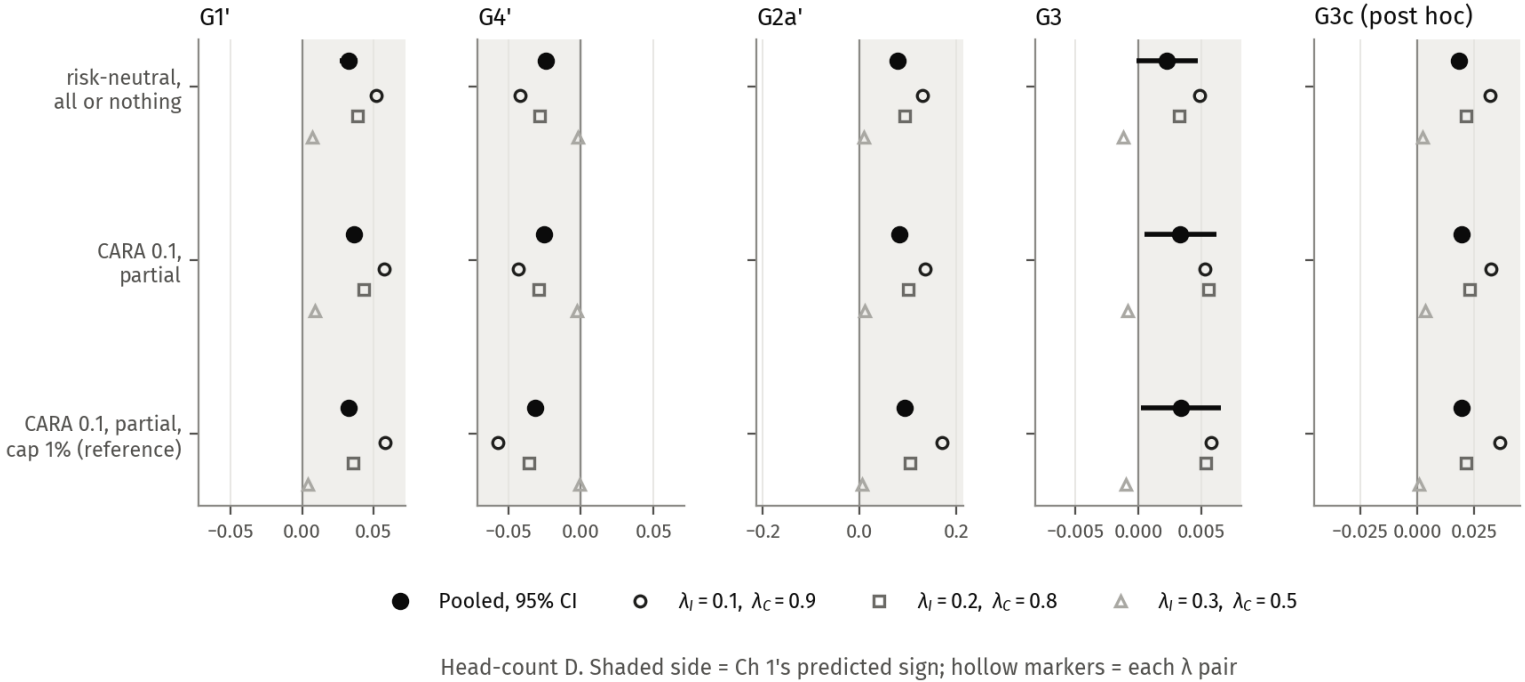
Technical versions of the exhibits, the pre-registered tests, channels, and the Celsius docking gap.

Who withdraws, by type and mix



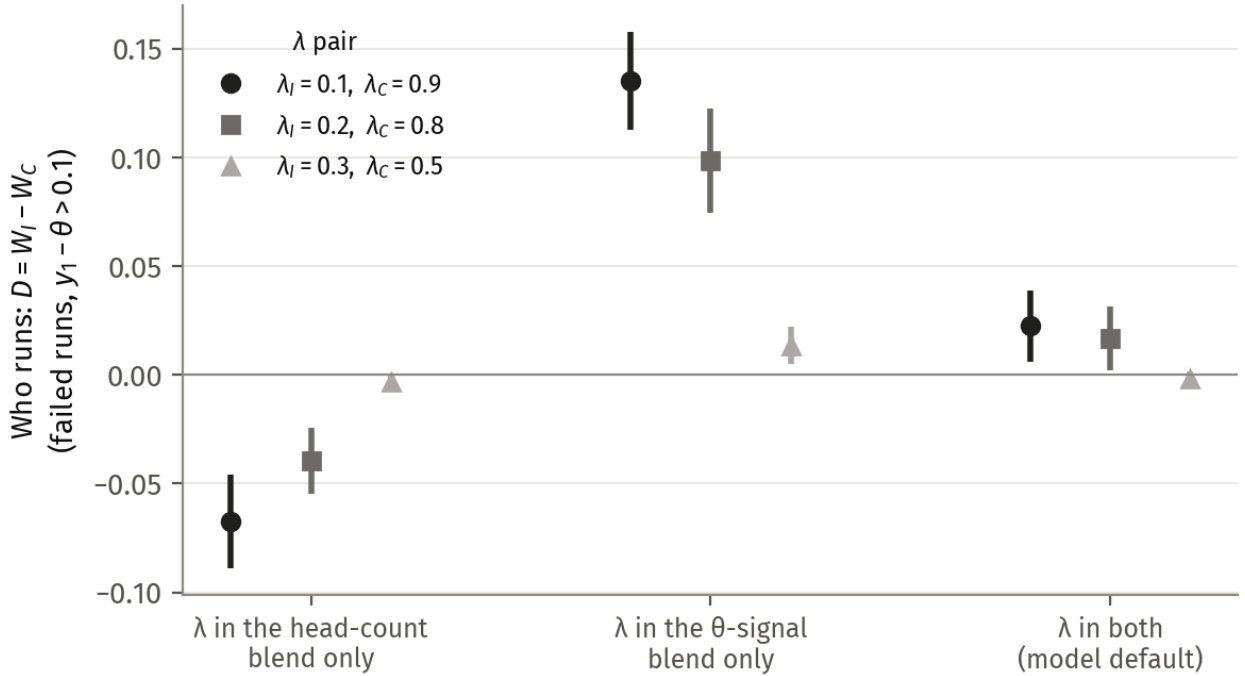
Share who withdraw by individualist share μ, one panel per λ pair; orange individualists, blue collectivists, with CIs.

The pre-registered tests, three versions of the bank



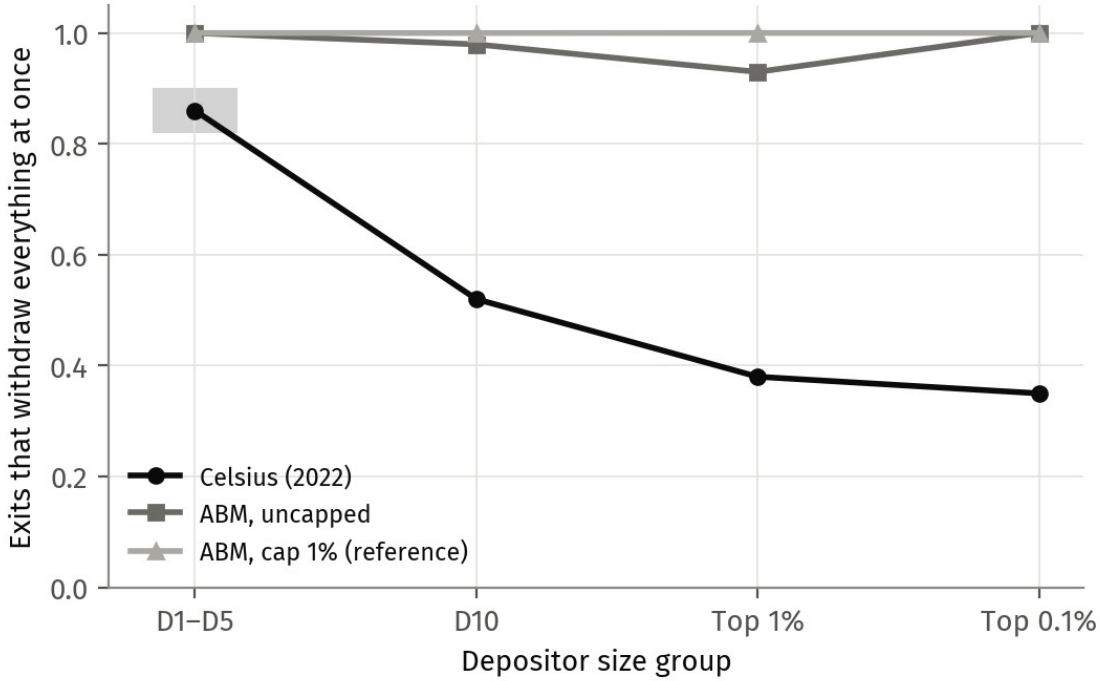
G1′, G4′, G2a′, G3 and G3c (post hoc). Rows: risk-neutral all-or-nothing, CARA partial, capped reference bank. Filled dot: pooled; hollow: each λ pair. Shaded: the theory's predicted sign.

Which channel carries culture



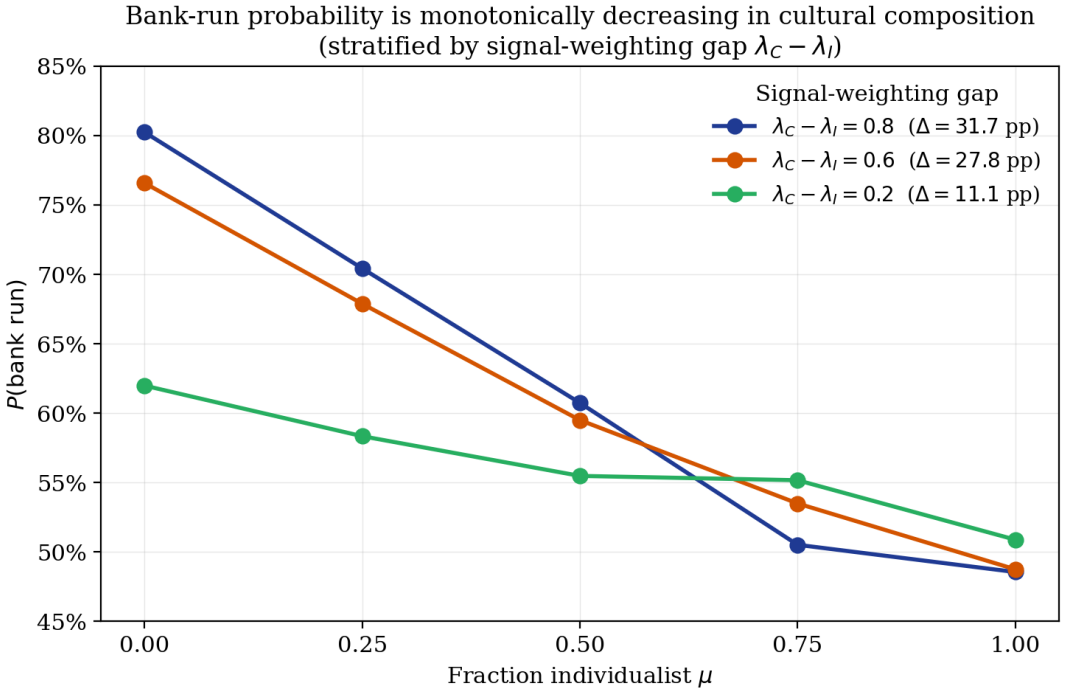
D in failed runs with optimistic news, when λ weights only the neighbour blend, only the news blend, or both (the model). θ-blend only +0.14; head-count only −0.07; both +0.02.

Matching Celsius: how large depositors exit



Share of exits that withdraw everything at once, by depositor size. Black: Celsius (band for D1–D5); greys: the model, uncapped and capped.

Earlier specification (vault bank): failure by mix



Vault bank, earlier specification (chapter Figure 1): probability of a run by share of individualists, three λ gaps. Also monotone.